DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 13

Python data analyst

Profession 13 of 10 in Development and IT.

What the profession means

A Python data analyst uses Python and pandas to clean, analyse and visualise data, and to automate reports that would take hours by hand.

What professionals do every day

Loads files, cleans them with pandas, computes summaries, produces charts and exports results, and schedules the scripts.

Why businesses hire them, and what problems they solve

Repeated reports are slow and error-prone by hand. A script runs the same steps correctly every time.

- Manual steps that differ each month.
- Data types that break calculations.
- Scripts without tests.

Skills you need

- Python basics.
- pandas: load, filter, group, merge.
- Plotting with matplotlib or similar.
- Testing and reproducible scripts.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Jupyter	Notebooks	Free	Free	jupyter.org
matplotlib	Charts	Free	Free	matplotlib.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write a script that loads a CSV and prints the summary.
Intermediate	Add an assert that checks a total.
Advanced	Export a chart as an image.

Projects by level

- **Beginner:** Monthly sales report automation.
- **Intermediate:** Survey data cleaning.

- **Advanced:** Customer summary script.

Client-ready portfolio project

A script that produces a monthly summary from a CSV, with asserts that check totals and an exported chart, for a fictional shop.

Worked example

Worked example. A pandas groupby gives revenue by channel; an assert checks the total equals the sum of the raw amounts before the report is written.

Assignment with full solution

ASSIGNMENT

Automated summary with checks

Business problem: A shop owner receives the same report by hand every month.

Steps

1. Load the orders file with pandas.
2. Group by channel and sum the amounts.
3. Assert that the grouped total equals the raw total.
4. Write the report to a file.

Expected output: A script, its checks and the report.

SOLUTION (ANSWER KEY)

PYTHON

```
import pandas as pd
orders = pd.DataFrame({'channel':['search','social','email','search','social'],'amount':
[1200,800,900,2500,1100]})
by_channel = orders.groupby('channel')['amount'].sum()
assert by_channel.sum() == orders['amount'].sum()
print(by_channel)
print('reconciliation passed')
```

Common mistakes

- Editing the data by hand between runs.
- No checks on totals.
- Hard-coded file paths.
- Scripts nobody else can run.

Freelancing opportunities

- Automated report scripts for small businesses.
- Data cleaning services.

- Briefs for 'Python', 'pandas', 'data automation'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- pandas documentation
- Python tutorial (official)

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.